

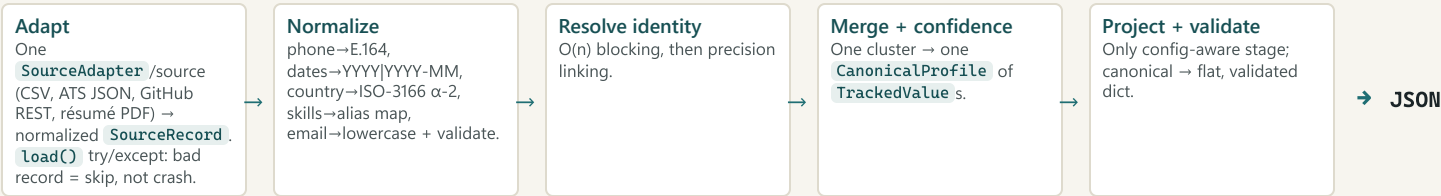
Candidate Profile Transformation Pipeline

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Goal. Ingest heterogeneous candidate records from multiple sources, resolve which belong to the same person, merge into one canonical profile per person with per-field provenance and confidence, and emit through a runtime-configurable projection.

- Never fabricate a value
- Never crash the batch – skip + log
- Deterministic & explainable

1. Pipeline



Threaded through every stage: a `RunReport` of skips / conflicts / assumptions / counts.

2. Canonical schema & normalized formats

CanonicalProfile: `candidate_id` (deterministic SHA1 of the strongest stable anchor — email → phone → name-key, so reruns are identical), `full_name`, `emails[]`, `phones[]` (• = confidence-sorted), `location`{city, region, country}, `links`{linkedin, github, portfolio, other[]}, `headline`, `skills[]`{name, confidence, sources[]}, `experience[]`, `education[]`, `repos[]`, `years_experience`, `overall_confidence`, `flags[]`. Every value is a **TrackedValue** {value, confidence, sources[], provenance[], competitors[]}.

Dates — `YYYY` or `YYYY-MM`, granularity preserved; "Present"/"" → None (ongoing); never fabricate a month.

Phones — E.164 via `phonenumbers`; missing country code + `--default-region` → applied & flagged; no region → keep raw, no guess.

Country — ISO-3166 alpha-2 via `pycountry`; unresolved → null (raw kept).

Skills — alias-mapped, preserving `+ / #` (C++/C#/.NET never collapse to "c"); unknown kept verbatim at lower confidence + flag.

Email — lowercased + shape-validated; a bad value is dropped, not the record.

3. Merge / conflict-resolution policy

- Match keys** (union — any shared key co-blocks): normalized email, `github_login`, `name_block_key` (sorted set of first letters of name tokens — survives initials/reordering, e.g. "Sri Krishna V" ≡ "V, Sri K."). Exact email/login links outright; otherwise name-alignment + shared phone ≥ threshold. No pairwise fuzzy match over all records — keeps it O(n).
- Single-valued winner** = highest source trust, decided by: `ATS 0.90 > CSV 0.80 > résumé 0.75 > GitHub 0.70`
Losing values are retained as `competitors` (never discarded) + `Flag(conflict_resolved)` + a `ConflictEntry` in the report.
- Multi-valued** (skills, emails, phones) = union + dedup, each value scored independently.

confidence = clamp01(base + corroboration - extraction - conflict) * recency

`base` = winner's trust · corroboration +0.05/extra agreeing source × independence (ATS↔CSV 0.5, any↔GitHub 1.0), capped +0.10 · extraction −0.10 for prose/heuristic values (GitHub bio, résumé, free-text location) · conflict −0.05 · recency decays only time-varying fields (company/title/location/headline) via `last_updated`.

Overall = weighted Σ over core fields — an absent field contributes 0:
`name .25 email .20 phone .15 company .15 title .15 location .10`

DEFERRED Under time pressure — all pluggable behind existing seams: live LinkedIn / recruiter-notes adapters · résumé experience/education parsing (lean scope today) · fuzzy/embedding **name & skill** matching (deterministic alias + blocking chosen for explainability) · embedding skill-similarity for the long tail (design noted in the README).

4. Runtime custom-output config

A JSON config is an ordered list of `FieldSpec`: `path` (output key), `from` (canonical source path — `field`, `field.sub`, `field[]`.sub; defaults to `path`), `type` (`string` | `string[]` | `number` | `object` | `object[]` — drives a dynamically built model), `required`, `normalize` (**assert-only**: verifies the value already matches E164/iso3166-a2/canonical form — a mismatch is treated as missing, the projector never recomputes, so double-normalization/drift can't happen), `on_missing` (`null` | `omit` | `error`), and `include_confidence` / `include_provenance`; plus global `on_missing` + `include_flags`.

The **Projector is the only config-aware component** — a hard boundary at **CanonicalProfile**; nothing upstream knows config exists. Output is validated by a Pydantic model built **at runtime** from the config (`create_model`); a profile that fails validation is skipped + logged and the batch continues. The default config reproduces the required output schema verbatim; a custom config renames fields, reshapes, or toggles confidence + provenance with **zero code change**.

5. Edge cases & deliberate descopes

HANDLED Company conflict — ATS *Stripe* vs CSV *Shopify*: trust picks the winner, losers kept as competitors, `conflict_penalty` + flag.

HANDLED Name variants, no shared email — "Sri Krishna V" / "Vijayarajan": blocking on `name_block_key` + `github_login` collapses them to one cluster; a GitHub record with no email/login and a non-aligning name stays an **orphan** (documented limitation).

HANDLED Garbage source — malformed CSV/JSON, corrupt PDF, or a scanned (no-text) résumé: adapter try/except → `RunReport` skip, other sources still emit.

HANDLED Phone without country code — `--default-region` applied + `assume:default_region` method + `Flag(assumed_region)`; with no region, the raw is kept and never guessed.

HANDLED Skill alias + unknown — `ReactJS` → `React`, C++/C#/.NET preserved intact; unknown kept verbatim @0.60 + flag.